IT4010 Research Project – 2025 July

Topic Assessment Form

Project ID : 25-26J-154

1. Topic (12 words max)

AI-Based Social Integration and Inclusion Platform for University Freshers

2. Research group the project belongs to

TIM - Technology Integration and Management

3. Specialization of the project belongs to

Information System Engineering(ISE)

4. If a continuation of a previous project:

Project ID	
Year	

5. Brief description of the research problem including references (200 – 500 words max) – references not included in word count.

University freshmen face growing challenges in social integration, academic adjustment, and mental health, with first-year retention rates dropping to 67% and social isolation as a key factor [1]. Nearly 48% of freshmen screen positive for anxiety or depression, yet only 22% access counseling services [2]. Traditional orientation and support systems fail to address these issues, especially for underrepresented students. Our AI-based platform uses adaptive profiling, behavioral analytics, and intelligent peer-matching to foster meaningful connections and early academic intervention. It offers conversational wellbeing support and personalized resources [3], providing a scalable, data-driven solution to improve student success and retention across domains.

- How does Conversational Support and Student Wellbeing Management help to solve the problem? In 2024, 38% of university students reported moderate to severe depression and 34% experienced anxiety, yet only a small portion accessed effective support [4]. Our AI-powered system addresses this by offering empathetic conversations, automated wellbeing FAQs, and personalized chat flows based on emotional cues. It escalates severe cases to human counselors and integrates student feedback to enhance engagement and support. This proactive, adaptive approach helps close mental health service gaps while promoting timely and stigma-free emotional care.
- How does Adaptive Student Profiling and Learning Persona Management help solve the problem? In 2024, nearly 46% of teachers reported that student engagement declined compared to pre-pandemic levels [5]. Traditional static learner profiles fail to address disengagement and diverse motivational drivers. By dynamically adapting student personas through behavioral, emotional, and social data, our system personalizes learning pathways in real-time, improving relevance and motivation. Continuous updates, social interaction analysis, and emotional

tracking help align support with each student's evolving context, addressing engagement gaps and enhancing learning outcomes through individualized, adaptive interventions.

- How does Intelligent Peer Matching and Relationship Evolution Management help solve the problem? University Freshers often struggle with social integration, as approximately 65 % report regular loneliness, leading to depression, poor academic engagement, and isolation [6]. Traditional onboarding programs lack dynamic matching mechanisms to address evolving emotional and social needs. This project tackles that gap by developing an AI-driven peer-matching and relationship evolution system designed to foster meaningful, adaptive connections and improve freshman wellbeing and retention.
- How does Social Inclusion Risk Detection and Early Intervention Management help to solve the problem? In 2024, 31% of university students reported moderate to high levels of loneliness and 25% experienced social anxiety, yet many did not seek help due to stigma or lack of awareness [7]. Our system addresses this by analyzing emotional and behavioral patterns to detect early signs of social withdrawal or stress. It offers tailored suggestions like connecting with peers and escalates serious cases to the conversational flow. This proactive, personalized approach supports student wellbeing by identifying and supporting silently struggling individuals before issues escalate.

References:

[1] A. Inscore, "The Healthy Minds Study," The Healthy Minds Network, 2024.

[2] National Student Clearinghouse Research Center, "Persistence and Retention," National Student Clearinghouse Research Center, 26 Jun 2025. [Online]. Available: https://nscresearchcenter.org/persistence-retention/. [Accessed 27 Jun 2025].

[3] NEWPORT INSTITUTE, "Depression in College Students: New Statistics and Research," NEWPORT INSTITUTE, 2024. [Online]. Available: https://www.newportinstitute.com/resources/mental-health/depression-on-college-campuses/?_gl=1*53k0sg*_up*MQ..*_ga*MTI1MTQyNzM1Ny4xNzUxMDMxNTEy*_ga_HMWTBRSNTM*czE3NTEwMzE1MTEkbzEkZzEkdDE3NTEwMzE1NzIkajYwJGwwJGgw. [Accessed 27 Jun 2025].

[4] EDUCAUSE, "2024 EDUCAUSE Horizon Report | Teaching and Learning Edition," EDUCAUSE, 13 May 2024. [Online]. Available: https://library.educause.edu/resources/2024/5/2024-educause-horizon-report-teaching-and-learning-edition. [Accessed 27 Jun 2025].

[5] E. K. Fittes, "Student Engagement in Learning Is Low. A Survey Offers Clues on How to Bring Them Back," EdWeek Market Brief, 24 Oct 2024. [Online]. Available: https://marketbrief.edweek.org/meeting-district-needs/student-engagement-in-learning-is-low-a-survey-offers-clues-on-how-to-bring-them-back/2024/10. [Accessed 27 Jun 2025].

[6] A. Eser, "Student Loneliness Statistics," ZIPDO, 30 May 2025. [Online]. Available: https://zipdo.co/student-loneliness-statistics/. [Accessed 27 Jun 2025].

[7] JMIR Publications, "Therapeutic Potential of Social Chatbots in Alleviating Loneliness and Social Anxiety: Quasi-Experimental Mixed Methods Study," Journal of Medical Internet Research, 14 Jan 2025. [Online]. Available: https://www.jmir.org/2025/1/e65589. [Accessed 27 Jun 2025].

6. Brief description of the nature of the solution including a conceptual diagram (250 words max)

An AI-based social integration and inclusion platform is proposed to address the challenges faced by university freshers. It is designed as a modular system consisting of four core components as Adaptive Student Profiling, Intelligent Peer Matching, Social Inclusion Risk Detection, and Conversational Support for Wellbeing.

- The platform begins with adaptive student profiling, which captures academic behavior, emotional states, and interaction patterns to form dynamic learning personas. These personas are continuously updated using AI-driven behavioral analytics.
- The intelligent peer matching component leverages compatibility metrics, such as shared interests, communication styles, and emotional profiles to suggest meaningful student connections. This addresses social isolation by promoting organic peer networks.
- The social inclusion risk detection framework uses real-time engagement data, sentiment analysis, and academic activity to identify at-risk students. Early warnings enable proactive interventions by university support staff.
- A conversational wellbeing support system offers emotional support via an AI-powered conversational support. It provides personalized strategies, and FAQ responses. In cases of severe emotional distress, the system intelligently escalates cases to human counselors.

These components provide a comprehensive solution that not only responds to students' challenges but also anticipates and mitigates them. The platform promotes emotional resilience, fosters inclusive peer networks, and supports students in building a strong sense of belonging during their transition to university life.

Figure 1: Conceptual Diagram

7. Brief description of specialized domain expertise, knowledge, and data requirements (300 words max)

This project operates at the intersection of educational technology, artificial intelligence (AI), behavioral science, and human-computer interaction. Specialized domain expertise is required in the areas of AI-driven adaptive learning, emotional analytics, decision modeling, and social network analysis. Understanding pedagogical theories, cognitive development, and student engagement metrics is also essential for developing meaningful learning personas and intervention strategies.

From a technical perspective, and sentiment behavioral, emotion detection is crucial to analyze student inputs, platform activity, and communication patterns. Knowledge of data visualization techniques supports the creation of intuitive dashboards for educators and students, while project management principles help design scalable wellbeing and support systems.

The system requires access to diverse and sensitive datasets. These include:

- Student behavioral data, such as login frequency, interaction patterns, and learning content preferences.
- Academic performance data, including assessments, quiz results, and peer evaluations.
- Student emotional data, derived from chats, journaling entries, or system interactions.
- Social interaction data, such as group participation, peer feedback, and network connectivity.

All data must be handled with strict adherence to data privacy, security, and ethical guidelines, especially when analyzing emotional or mental health-related signals. Anonymization and secure data storage protocols are vital.

Ultimately, the project demands a multidisciplinary blend of educational psychology, AI, data science, and user experience design to ensure the system delivers accurate insights, effective peer relationships, early interventions, and personalized support in a trustworthy and student-centric manner.

8. Objectives and Novelty

Main Objective **AI-Based Social Integration and Inclusion Platform for University Freshers**: To develop an AI-based platform that promotes social integration and emotional wellbeing among university freshers by providing personalized support through adaptive profiling, intelligent peer matching, early risk detection, and emotionally responsive conversational systems. The platform aims to reduce loneliness, enhance student engagement, and ensure inclusive support during the critical transition into university life.			
Member Name with Registration No	Sub Objective	Tasks	Novelty
H. K. Liyanage IT22133540	Conversational Support and Student Wellbeing Management	**Developing AI Conversations to Provide Emotional Support:** Designed conversational flows that respond empathetically to students' emotional states, offering encouragement, coping tips, and mental wellness guidance. **Creating Auto Replies for Frequently Asked Questions:** Implemented automated responses to common queries related to student life, stress management, time handling, and campus resources. **Building Personalized Chat Flows for Each Student:**	The management offers more than just automated replies. It analyzes emotional cues, provides tailored support strategies, and helps students build emotional awareness. It includes tools like journaling with AI feedback and automatically escalates serious concerns, ensuring students receive the support they need at the right time.

		Structured individual chat experiences based on emotional cues, behavioral history, and previously detected wellbeing patterns. **Escalating Serious Cases to Human Counselors:** Established a decision-making process that automatically involves human intervention (e.g., referring students to counseling services) when signs of serious emotional distress are detected. **Monitoring Student Wellbeing Trends:** Enabled continuous tracking of student sentiment and emotional patterns through journaling, interaction history, and mood inputs. **Getting Student Feedback on the Conversation Platform:** Enabled the collection and analysis of student feedback on the platform's responses, usability, and emotional support effectiveness to improve conversational quality and user satisfaction continuously.	
H. N. Pathiraja IT22142214	Adaptive Student Profiling and Learning Persona Management	**Integrating AI Models to Generate Student Personas:** Use AI models or APIs to analyze behavioral, emotional, and motivational data, creating adaptive student	The management uses AI to monitor behavioral and emotional data to generate a unique learning persona that evolves over time. It creates a cognitive and emotional profile for

		personas that evolve with system interactions. **Enabling Continuous Updates of Student Profiles in Real-Time:** Develop a system for real-time or periodic updates of student personas by capturing platform interactions, emotional inputs, and academic engagement to refine profiles dynamically. **Analyzing Platform Usage and Behavioral Patterns:** Implement data collection to monitor student activity such as navigation, usage frequency, and interactions and analyze it to track engagement trends and evolving learning preferences. **Mapping Emotional and Motivational Indicators:** Use sentiment analysis and engagement scoring on structured inputs to assess emotional and motivational states, enhancing the cognitive-emotional depth of each student persona. **Applying Decision Modeling Techniques for Adaptive Personalization:** Use decision modeling to guide persona evolution and trigger personalized	each student, allowing the system to personalize the learning experience at a deeper level than fixed profile systems.

		support based on behavioral and emotional patterns. **Integrating Social Interaction Analytics:** Apply social network analytics to peer interactions and group activities to understand social learning behavior and refine student personas.	
W. A. N. P. Fonseka IT22240842	Intelligent Peer Matching and Relationship Evolution Management	**Developing Compatibility-Based Matching Algorithms:** Design and implement AI algorithms to match freshers based on academic background, shared interests, emotional traits, and communication preferences. **Group Compatibility Scoring System:** Build a scoring system that evaluates peer group health based on emotional diversity, participation level, and academic alignment. **Relationship Evolution and Re-Matching Logic:** Enable the system to reassign or recommend new peers or mentors when student compatibility weakens or feedback indicates low engagement. **Feedback Loop and Social Graph Visualization:** Integrate feedback collection after peer interactions and use social network	The system goes beyond traditional static groupings, by using AI to adapt peer relationships over time. It proactively restructures social connections based on real-time behavior, communication patterns, and emotional states. This dynamic and responsive peer networking approach ensures that students are consistently supported and connected to those best suited for their growth and well-being.

		visualizations to track student connectivity and relationship strength. **Integrating Gamified Friend-Matching Experience:** Add gamified features such as badges for "Best Peer Match", "Supportive Teammate" for interaction to increase student participation and motivation.	
M. P. H. Poornima IT22917584	Social Inclusion Risk Detection and Early Intervention Management	**Detecting At-Risk Students Using Behavioral and Emotional Data:** Used AI models to analyze participation patterns, sentiment trends, and emotional cues to identify students at risk of isolation or burnout. **Triggering Personalized Early Interventions:** Built logic to send timely suggestions such as check-ins or peer connections when risk is detected. **Creating Visual Risk Dashboards:** Designed visualizations to display student risk levels in a clear and actionable manner for staff. **Monitoring Usability and Engagement:** Applied usability analytics to ensure students interact with the system effectively and respond to interventions.	This system introduces a multi-dimensional risk detection framework that goes beyond basic academic indicators by using AI to analyze emotions and behaviors. It identifies students facing social isolation, emotional stress, or early signs of disengagement. Unlike one size fits all alerts, it differentiates between risk types, recognizing students who silently struggle but are still open to help. The system offers personalized suggestions such as peer prompts or wellbeing resources and escalates critical cases to conversational flow. This balanced, context-aware approach enables earlier, more effective support for student wellbeing.

9. Individual component description of how it is complied with the specialization.

Member Name with Registration No	Description
H. K. Liyanage IT22133540	**Conversational Support and Student Wellbeing Management:** This management aligns with Information Systems Engineering by utilizing decision modeling to guide personalized wellbeing pathways based on students' emotional and behavioral data. Human-computer interaction is applied to design emotionally responsive conversational interfaces that ensure comfortable and effective user engagement. Project management principles are used to organize, implement, and monitor wellbeing initiatives across the platform. Visualizations are employed to represent emotional trends in an accessible format, supporting better understanding and timely interventions.
H. N. Pathiraja IT22142214	**Adaptive Student Profiling and Learning Persona Management:** This component aligns with Information Systems Engineering by applying decision modeling to create and update adaptive student profiles based on behavior. Visualizations are used to represent the learning persona data in a user-friendly way. Technology adoption and diffusion are considered when implementing and refining the profiling system. Social network analytics helps to understand student behavior in relation to peers.
W. A. N. P. Fonseka IT22240842	**Intelligent Peer Matching and Relationship Evolution Management:** This component integrates Information Systems Engineering areas such as decision modeling, which is used to drive the matching logic based on multi-dimensional compatibility factors (academic, emotional, social). It applies social network analytics to identify relationship strengths, group engagement levels, and structural weaknesses within student communities. Additionally, usability and feedback analytics ensure student interactions with the system are tracked and used to enhance future recommendations.

M. P. H. Poornima IT22917584	**Social Inclusion Risk Detection and Early Intervention Management:** This part matches the Information Systems Engineering field because it uses AI and data to find students who might feel lonely or stressed. It checks how students behave on the platform and shows their risk levels using simple charts. The system suggests helpful actions early, like talking to friends. It also makes sure students are comfortable using the system by tracking their responses. Overall, it combines technology and emotional understanding to support students better, which is an important part of ISE.

10. Supervisor details

	Title	First Name	Last Name	Signature
Supervisor	Ms.	Narmada	Gamage	
Co-Supervisor	Ms.	Chathurangika	Kahandawaarachchi	
External Supervisor				
Summary of external supervisor's (if any) experience and expertise				

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IT4010 Research Project – 2025 July

Topic Assessment Form

This part is to be filled by the Topic Screening Staff members.

a) Does the chosen research topic possess a comprehensive scope suitable for a final-year project?

Yes	✓	No	

b) Does the proposed topic exhibit novelty?

Yes	✓	No	

c) Do you believe they have the capability to successfully execute the proposed project?

Yes	✓	No	

d) Do the proposed sub-objectives reflect the students' areas of specialization?

Yes	✓	No	

e) Supervisor's Evaluation and Recommendation for the Research topic:

Students required to access freshers data.
Recommended to initiate data collection as soon as possible & identify other required external support as well.
scope is sufficient.

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IT4010 Research Project – 2025 July Topic Assessment Form

Acceptable: Mark/Select as necessary

Topic Assessment Accepted	
Topic Assessment Accepted with minor changes*	
Topic Assessment to be Resubmitted with major changes*	
Topic Assessment Rejected. Topic must be changed	

* Detailed comments given below

Comments

Staff Member's Name	Signature

*__Important__:

1. According to the comments given by the evaluator, make the necessary modifications and get the approval by the **Evaluator**.
2. If the project topic is rejected, identify a new topic, and request the RP Team for a new topic assessment.